

Playwright Interview Questions with Detailed Answers (Advanced 2026 Edition)

1. What makes Playwright different from Selenium?

Playwright is a modern end-to-end automation framework developed by Microsoft.

Key differences:

- Built-in auto-waiting mechanism (reduces flaky tests).
- Supports Chromium, Firefox, and WebKit with a single API.
- No need for separate browser drivers.
- Native parallel execution.
- Better handling of modern web apps (SPA, Shadow DOM).
- Built-in tracing, video recording, and screenshots.

In contrast, Selenium requires WebDriver setup and explicit waits more frequently.

2. Explain Auto-Waiting in Playwright.

Playwright automatically waits for elements to be:

- Attached to the DOM
- Visible
- Stable (not animating)
- Enabled and actionable

Before performing actions like `click()`, `fill()`, or `hover()`.

This eliminates the need for `Thread.sleep()` or explicit waits in most cases, making tests stable and reliable.

3. Difference between `page.locator()` and `page.$()`?

`page.locator()`:

- Recommended approach.
- Supports auto-waiting and retry logic.
- Lazy evaluation (evaluated when action is performed).
- Works well with assertions.

`page.$()`:

- Returns element handle immediately.
- No retry or auto-wait.
- Can lead to flaky tests if element is not ready.

4. How do you handle Iframes in Playwright?

Playwright provides `frameLocator()` for interacting with iframes.

Example:

```
const frame = page.frameLocator('#frameId');  
await frame.locator('button').click();
```

This ensures reliable interaction inside embedded frames.

5. How do you validate API responses in UI automation?

Use `page.waitForResponse()` to capture network calls.

Example:

```
const response = await page.waitForResponse('**/api/login');  
expect(response.status()).toBe(200);
```

You can validate:

- Status codes
- Response body
- Headers
- Response time

This helps combine UI + API validation in a single test.

6. Explain Parallel Execution in Playwright.

Playwright Test Runner runs tests in parallel by default using worker processes.

Configuration:

workers: 4

Fully isolated browser contexts ensure no shared session or cookie conflicts.

7. How do you implement Page Object Model (POM)?

Create separate classes for each page.

Example:

```
class LoginPage {  
  constructor(page){  
    this.page = page;  
    this.username = page.locator('#username');  
  }  
  async login(user, pass){  
    await this.username.fill(user);  
  }  
}
```

This improves maintainability and reusability.

8. How to capture Screenshots & Videos?

In playwright.config.js:

```
use: {  
  screenshot: 'only-on-failure',  
  video: 'on-first-retry',  
  trace: 'on-first-retry'  
}
```

This helps debugging failed executions.

9. How to integrate Playwright in CI/CD?

Steps:

1. Install dependencies (npm install).
 2. Add command: npx playwright test.
 3. Store reports as artifacts.
 4. Run in headless mode.
- Supported platforms: Jenkins, GitHub Actions, GitLab CI.

10. How to handle Multi-User Scenarios?

Use multiple browser contexts:

```
const user1 = await browser.newContext();  
const user2 = await browser.newContext();
```

Each context simulates a separate user session with isolated cookies and storage.

11. How to test File Upload and Download?

Upload:

```
await page.setInputFiles('#upload', 'file.pdf');
```

Download:

```
const download = await page.waitForEvent('download');  
await download.saveAs('path/file.pdf');
```

12. How do you reduce flaky tests?

Best Practices:

- Use locators instead of element handles.
- Avoid hard waits.
- Use assertions with auto-wait.
- Use retries in configuration.

- Stabilize test data and environment.

13. How do you generate Reports?

Use built-in HTML reporter:

```
reporter: [['html', { open: 'never' }]]
```

Generates detailed execution report with steps and traces.

14. How do you test Shadow DOM?

Playwright automatically pierces Shadow DOM using locators.

Example:

```
await page.locator('custom-element >> text=Submit').click();
```

15. Real-world Example Question: How will you automate Login with OTP?

Strategy:

- Enter credentials.
- Wait for OTP API response.
- Fetch OTP from test environment (mock/email/db).
- Validate successful login.
- Add negative tests for invalid/expired OTP.